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Localized Algorithm for Segregation of Critical/Non-critical Nodes in Mobile Ad Hoc and Sensor Networks

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Abstract

Timely segregation of connectivity-centric critical/non-critical nodes is extremely crucial in mobile ad hoc and sensor networks to assess network vulnerabilities against critical node failures and provide precautionary means for survivability. This paper presents a localized algorithm for segregation of critical/non-critical nodes (LASCNN) that opts to distinguish critical/non-critical nodes to the network connectivity based on limited topology information. Each node establishes and maintains a k-hop connection list and employ LASCNN to determine whether it is critical/non-critical. Based on the list, LASCNN marks a node as critical if its k-hop neighbor's become disconnected without the node, non-critical otherwise. Simulation experiments demonstrate the scalability of LASCNN and shows the performance is quite competitive compared to a scheme with global network information. The accuracy of LASCNN in determining critical nodes is 87% (1-hop) and 93% (2-hop) and non-critical nodes 91% (1-hop) and 93% (2-hop).

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1. Introduction

Mobile ad hoc and sensor networks (MAHSNs) are captivating significant attention from research community because of their suitability for enormous range of applications. Nodes in these networks are expected to establish and maintain a connected wireless network and operate autonomously in unattended setups. Due to ad hoc nature, limited energy and transmission range nodes have to rely on multihop communication i.e., a node forwards a message to the next node on the path towards the destination. The nodes organize themselves because there is no central entity to manage the network. Therefore, maintaining network connectivity is extremely desirable in order for the network to stay functional**.**

Nonetheless, random deployment, frequent mobility and network dynamics (e.g., sleep/wakeup) inherently introduce critical i.e., cut vertex nodes to the network connectivity. Nodes in these networks are

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